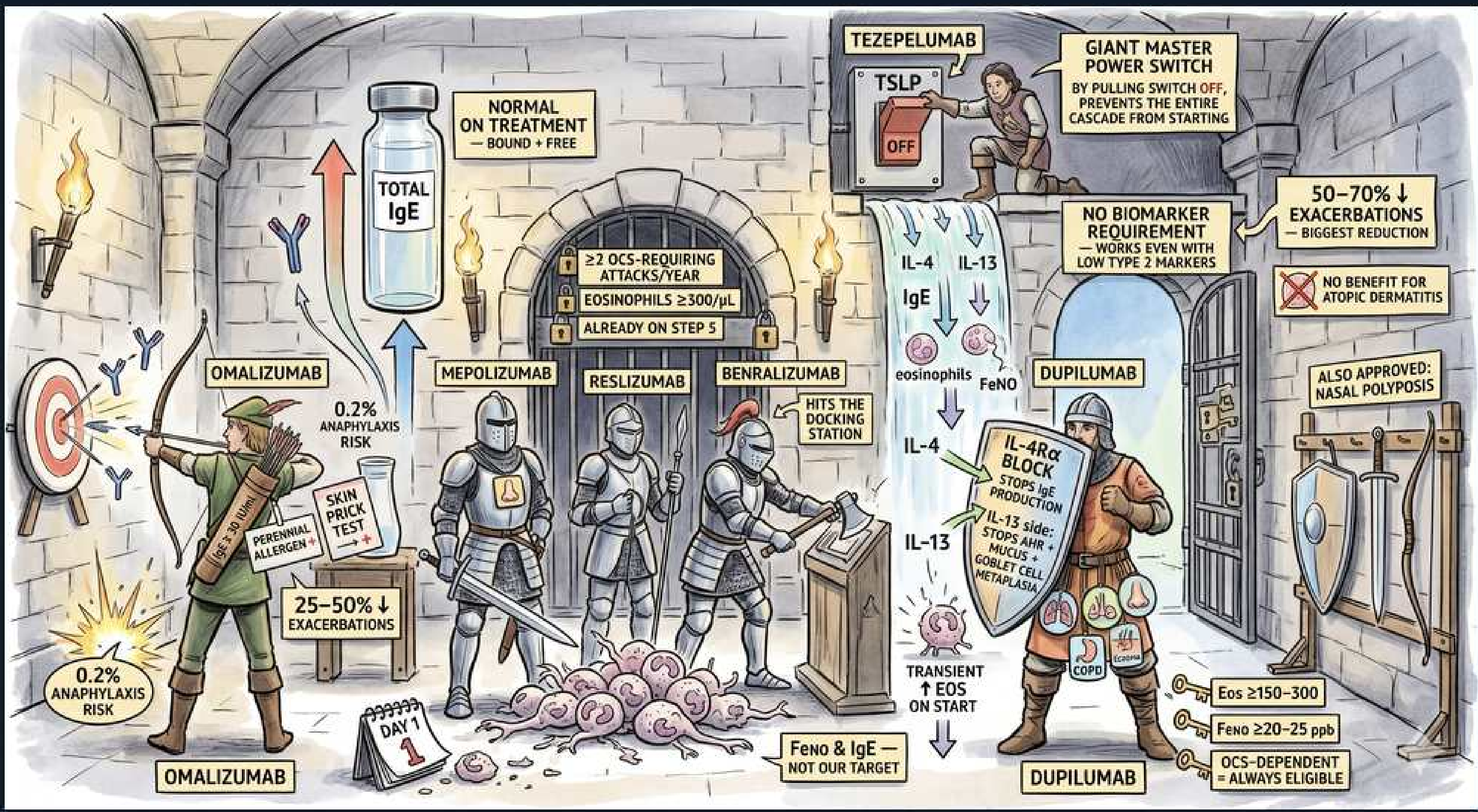
Asthma — Type 2 Biologics Castle



1. Tezepelumab master switch (anti-TSLP)

WHAT YOU SEE

A 'TEZEPELUMAB' figure flipping a 'TSLP' master power switch to OFF.

WHAT IT ENCODES

Tezepelumab blocks TSLP, an epithelial alarmin released at the very top of the cascade in response to allergens, viruses, and pollutants. By switching off this upstream master signal it suppresses the entire downstream type-2 axis (IL-4/IL-5/IL-13) — making it the broadest biologic, effective across phenotypes including some non-type-2 disease.

BOARD TRAP

Tezepelumab works even with LOW type-2 biomarkers — it is the one biologic with NO eosinophil/FeNO/IgE threshold requirement, so it's the answer when biomarkers are low but exacerbations persist.

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2. No biomarker requirement

WHAT YOU SEE

A sign 'NO BIOMARKER REQUIREMENT — works even with low type 2 markers'.

WHAT IT ENCODES

Because it acts upstream at the alarmin level, tezepelumab is usable regardless of eosinophil count or FeNO — the only severe-asthma biologic without a biomarker gate. Every OTHER biologic (anti-IgE, anti-IL5, anti-IL4R) requires an elevated selecting biomarker.

BOARD TRAP

The eosinophil/IgE-targeted biologics each require their specific biomarker to be elevated; only tezepelumab is biomarker-agnostic.

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3. Total IgE arrow (omalizumab target)

WHAT YOU SEE

A vial 'TOTAL IgE — bound + free' with an arrow.

WHAT IT ENCODES

Omalizumab dosing is calculated from baseline TOTAL serum IgE (bound + free) and body weight, using a dosing table. After it starts, measured total IgE rises (drug-IgE complexes) so total IgE can no longer be used to monitor therapy — dose stays fixed to the pre-treatment value.

BOARD TRAP

Don't recheck total IgE to titrate omalizumab — levels rise artifactually on therapy; dosing is set from the pretreatment IgE and weight only.

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4. Omalizumab archer (anti-IgE)

WHAT YOU SEE

An archer 'OMALIZUMAB' hitting a target with '0.2% anaphylaxis risk' and 'perennial allergen / skin-prick'.

WHAT IT ENCODES

Omalizumab (anti-IgE) binds free IgE, lowering it and downregulating mast-cell/basophil FcεRI receptors. It is indicated for moderate-severe ALLERGIC asthma with elevated IgE and proven perennial aeroallergen sensitization (positive skin-prick or specific IgE). It also reduces nasal polyps and chronic urticaria.

BOARD TRAP

Omalizumab carries a ~0.1-0.2% anaphylaxis risk (boxed warning) — observe after injection and prescribe an epinephrine auto-injector; it requires proven allergic sensitization, so it is NOT for non-allergic eosinophilic asthma.

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5. Mepolizumab knight (anti-IL-5)

WHAT YOU SEE

A knight 'MEPOLIZUMAB' among armored figures, eosinophils on the ground.

WHAT IT ENCODES

Mepolizumab is a subcutaneous monoclonal antibody that binds circulating IL-5, the key cytokine for eosinophil maturation, recruitment, and survival — depleting blood/airway eosinophils and cutting exacerbations ~50%. It is also approved for EGPA, hypereosinophilic syndrome, and CRSwNP.

BOARD TRAP

Anti-IL-5 agents need elevated blood eosinophils (typically ≥150 at initiation or ≥300 historically) — they are useless in non-eosinophilic asthma, so a low eosinophil count rules them out.

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6. Reslizumab knight (anti-IL-5)

WHAT YOU SEE

A knight 'RESLIZUMAB' beside mepolizumab.

WHAT IT ENCODES

Reslizumab is an anti-IL-5 antibody given INTRAVENOUSLY and dosed by weight (vs subcutaneous fixed-dose mepolizumab) — the IV route is its distinguishing feature and the reason it carries an anaphylaxis boxed warning requiring in-clinic administration.

BOARD TRAP

Reslizumab is the IV, weight-based anti-IL-5 (with anaphylaxis risk) — mepolizumab and benralizumab are subcutaneous; matching drug to route/dosing is a board distinction.

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7. Benralizumab knight (anti-IL-5Rα)

WHAT YOU SEE

A knight 'BENRALIZUMAB' 'hits the docking station'.

WHAT IT ENCODES

Benralizumab binds the IL-5 RECEPTOR alpha on eosinophils (not the cytokine), recruiting NK cells to trigger antibody-dependent cell-mediated cytotoxicity (ADCC) — causing near-complete, direct eosinophil DEPLETION. After loading it is dosed every 8 weeks, a convenience advantage.

BOARD TRAP

Benralizumab targets the IL-5 RECEPTOR (causing ADCC-mediated eosinophil killing); mepolizumab/reslizumab target the IL-5 LIGAND — receptor vs ligand is the testable mechanistic difference.

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8. IL-4 / IL-13 cytokines

WHAT YOU SEE

Labeled streams 'IL-4' and 'IL-13' feeding downstream targets.

WHAT IT ENCODES

IL-4 and IL-13 are the central type-2 effector cytokines: IL-4 drives Th2 differentiation and IgE class-switching; IL-13 drives IgE switching, goblet-cell mucus, airway hyperresponsiveness, and FeNO production. They signal through a shared IL-4Rα subunit — the target of dupilumab.

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9. Dupilumab (anti-IL-4Rα)

WHAT YOU SEE

A figure 'DUPILUMAB' with 'IL-4Rα block' stopping IL-4 and IL-13.

WHAT IT ENCODES

Dupilumab blocks the SHARED IL-4 receptor alpha subunit, simultaneously inhibiting BOTH IL-4 and IL-13 signaling — so it lowers FeNO, IgE, mucus, and exacerbations. Its broad downstream reach also makes it effective in atopic dermatitis, CRSwNP, eosinophilic esophagitis, and prurigo nodularis.

BOARD TRAP

Dupilumab may cause a transient eosinophil RISE on initiation (eosinophils can't migrate into tissue but keep being made) — usually benign, but monitor; rarely unmasks EGPA/hypereosinophilia.

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10. Dupilumab eligibility (Eos / OCS)

WHAT YOU SEE

Tags 'Eos ≥150-300', 'OCS-dependent = always eligible'.

WHAT IT ENCODES

Dupilumab is indicated for the eosinophilic phenotype (blood eos ≥150-300) OR elevated FeNO, AND it is appropriate for ANY oral-corticosteroid-dependent severe asthmatic regardless of baseline eosinophils — its OCS-sparing effect is a major advantage.

BOARD TRAP

OCS-dependence makes a patient eligible for dupilumab even if biomarkers look modest — chronic steroids suppress the eosinophil count, so don't exclude these patients on a 'low' eos value.

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11. FeNO biomarker

WHAT YOU SEE

A 'FeNO' / 'eosinophils' biomarker readout near the cascade.

WHAT IT ENCODES

FeNO (fractional exhaled nitric oxide) is generated by IL-13-driven epithelial iNOS, so it directly reflects type-2/IL-13 airway inflammation and helps select FeNO/eosinophil-responsive biologics (dupilumab, anti-IL5) and predict steroid response.

BOARD TRAP

FeNO and IgE are biomarkers/SELECTORS, not therapeutic targets — no asthma drug 'lowers FeNO' as its mechanism; FeNO just falls as inflammation is treated.

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12. Nasal polyposis FDA approval

WHAT YOU SEE

A sign 'also FDA approved: nasal polyposis' with 'no benefit for atopic dermatitis' note.

WHAT IT ENCODES

Type-2 biologics share comorbidity approvals: dupilumab, omalizumab, mepolizumab, and benralizumab are all approved for chronic rhinosinusitis with nasal polyps (CRSwNP), reflecting unified airway type-2 disease — a reason to favor a biologic that covers the patient's specific comorbidity.

BOARD TRAP

Dupilumab — NOT the anti-IL-5 agents — is the one also approved for ATOPIC DERMATITIS (and eosinophilic esophagitis); matching biologic to extra-pulmonary comorbidity is a favorite board question.
